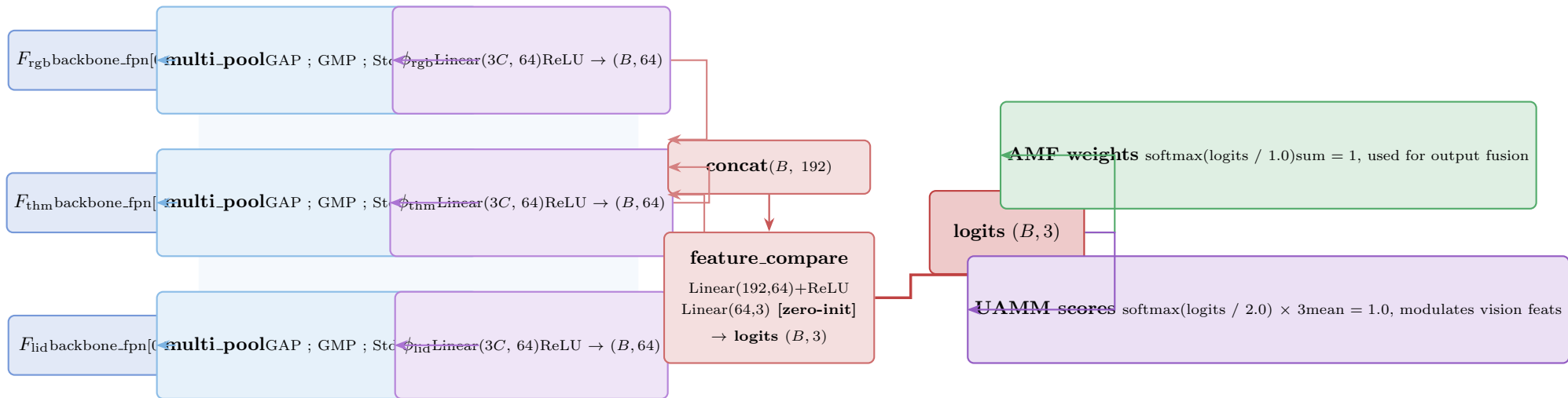


## CrossModalFusionHead V2 — Phase 2

Feature-level path



### Why per-modality compress?

P9 shared: modality cues averaged away

P10/P11 separate Linear per modality:

- RGB learns brightness/color cues
- Thermal learns contrast/saturation
- LiDAR learns spatial density

**multi\_pool** = **GAP** + **GMP** + **Std**:

GAP: mean signal; GMP: peak signal

Std: texture/noise level

### Zero-init strategy:

$\text{feature\_compare}[-1] = 0$

$\Rightarrow \text{logits}_{\text{init}} = 0$

$\Rightarrow \text{AMF init} = [1/3, 1/3, 1/3]$

Oracle KL loss gradually  
differentiates gating weights